



TARGET

JKBOSE CLASS-11TH

Structural Organization in
Animals

One Shot

Zoology

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Topics To Be Covered

Morphology ✓

Anatomy ✓



Digestive System ✓

Respiratory System ✓

Blood Vascular System ✓

Excretory System ✓



Nervous System ✓

Reproductive System ✓

Frog

↳ Common Species *Rana tigrana*

Frog = Similarities with Human
- Understand.

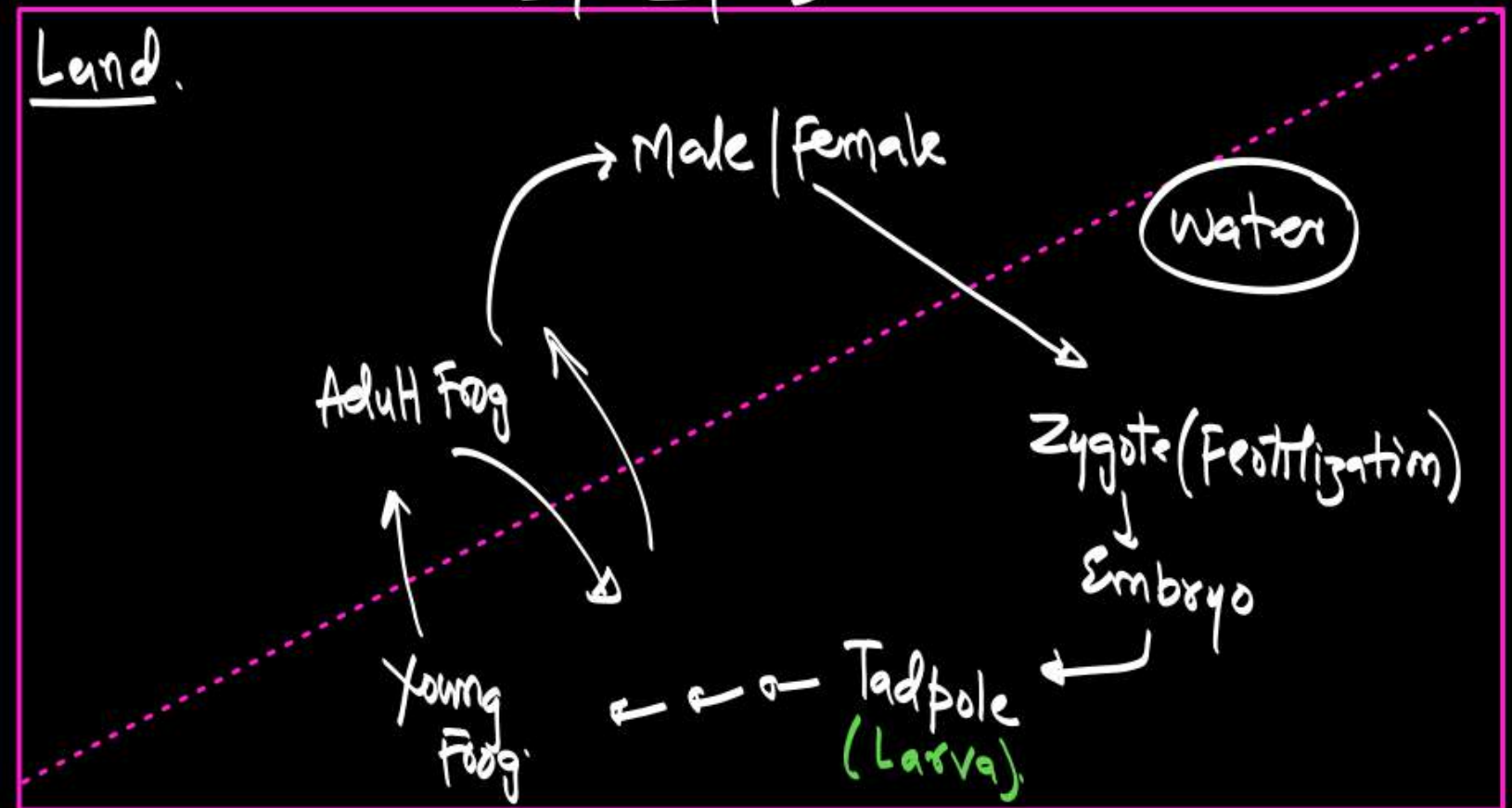
Frog (Rana tigrana)

- Kingdom - Animalia
- Phylum - Chordata
- Class - Amphibia
- Order - Anura (without tail)
- Family - Ranidae (Frogs)
- Genus - Rana
- Species - tigrana

→ Frog is an Amphibian
 ↓ ↓
 Dual Bios = life

→ Frog has Dual Nature of Life
 (Survive in H₂O as well as on Land)
 (- Aquatic, terrestrial)

(Life Cycle)

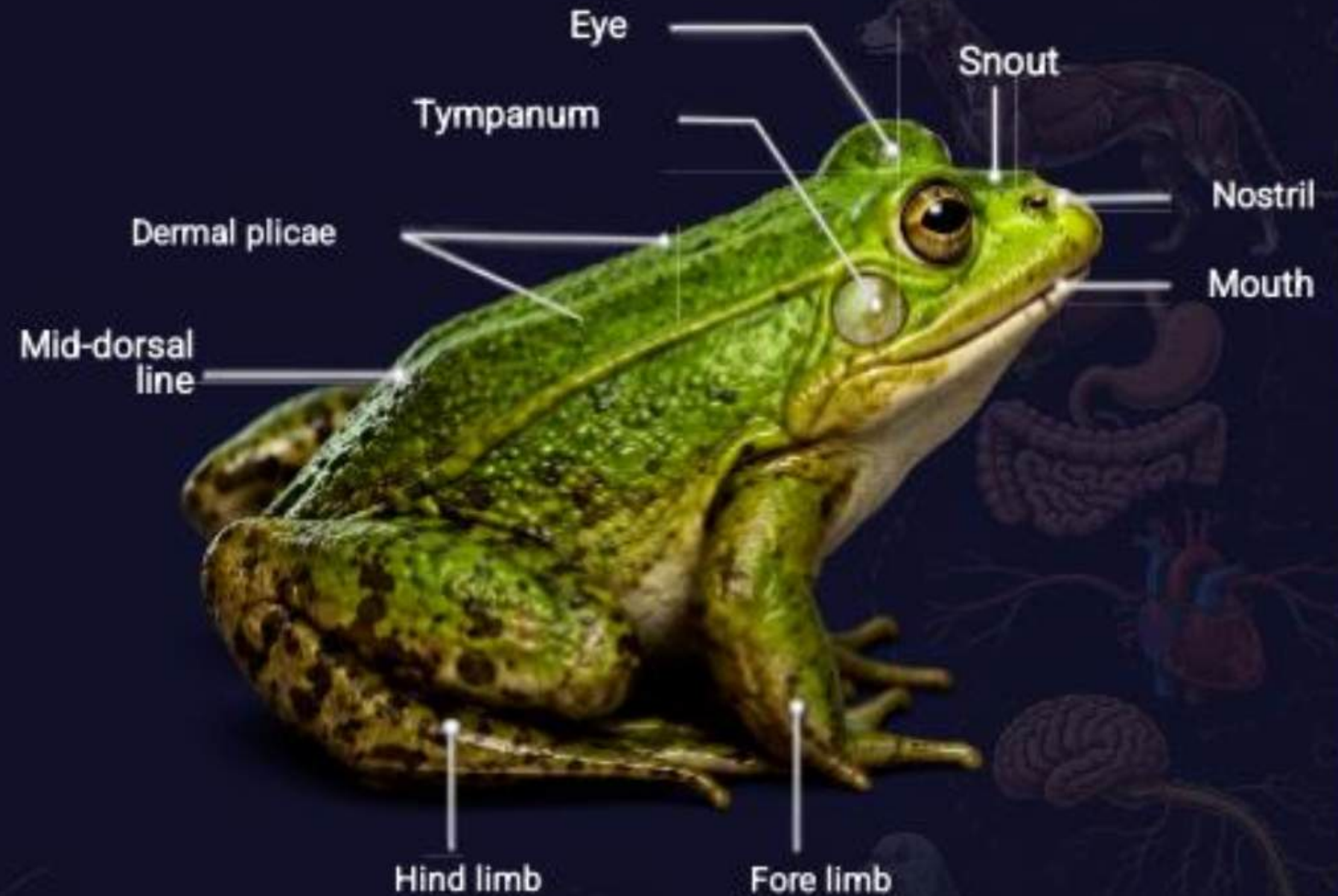


FROG



Classification:

- **Phylum:** Chordata
- **Subphylum:** vertebrata
- **Section:** Gnathostomata
- **Superclass:** Tetrapoda
- **Class:** Amphibia
- **Family:** Ranidae
- **Genus:** Rana
- **Species:** tigrina



GENERAL CHARACTERISTICS

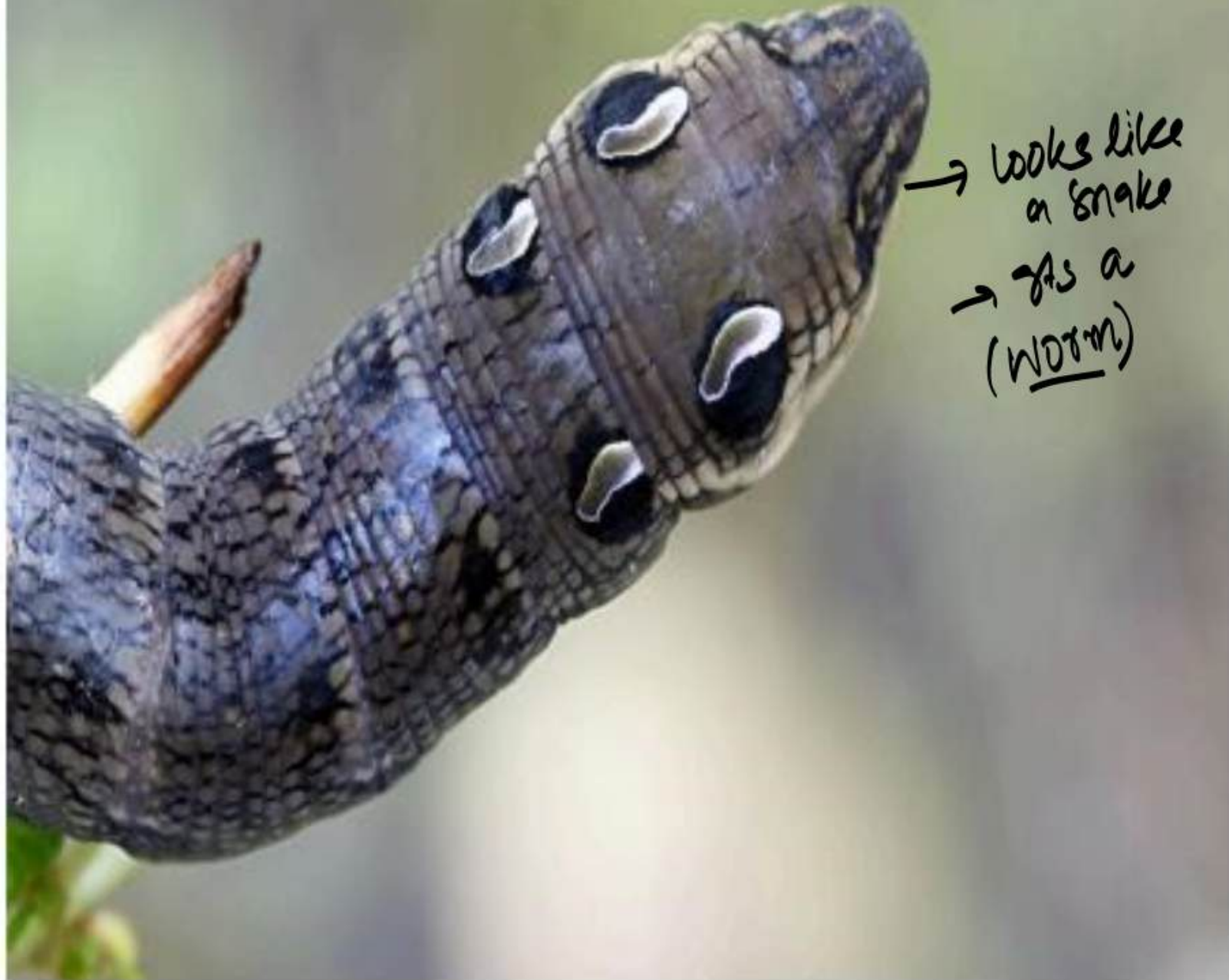


→ Body temp changes According Environment

- They are cold blooded or poikilotherms. Their body temperature is not constant.
- They have the ability to change the color to hide them from their enemies (camouflage). This protective coloration is called mimicry.
- During peak summer and winter they take shelter in deep burrows to protect them from extreme heat and cold.
- This is known as summer sleep (aestivation) and winter sleep (hibernation) respectively.
- Locomotion by leaping (1.5 to 2 m), forelimb have shock absorber, hindlimb is webbed.

Jumping

Mimicry - To look like Another living Being.



Camouflage (to Blend with Environment)

MORPHOLOGY

→ External characters
→ Mucous Gland

- Frogs have mucus present on their skin that makes it slippery and smooth. A moist condition is always maintained on skin.
- Generally, their body's dorsal side is olive green in color and has irregular dark spots.
- On the other hand, there is a uniform pale yellow color on the ventral side.
- The intake of water in frogs occurs as absorption through skin.
- Body of a frog is divisible into head and trunk.
- A neck and tail are absent.
- A pair of nostrils is found above the mouth.
- Eyes are bulged and covered by a nictitating membrane. (3rd eyelid).



- (Ear Drum)





— Male Frogs have

(a) Vocal sacs

(b) Copulatory pad

(c) Both (A) and (B) ✓

(d) None of these

This gives them Sexual Dimorphism

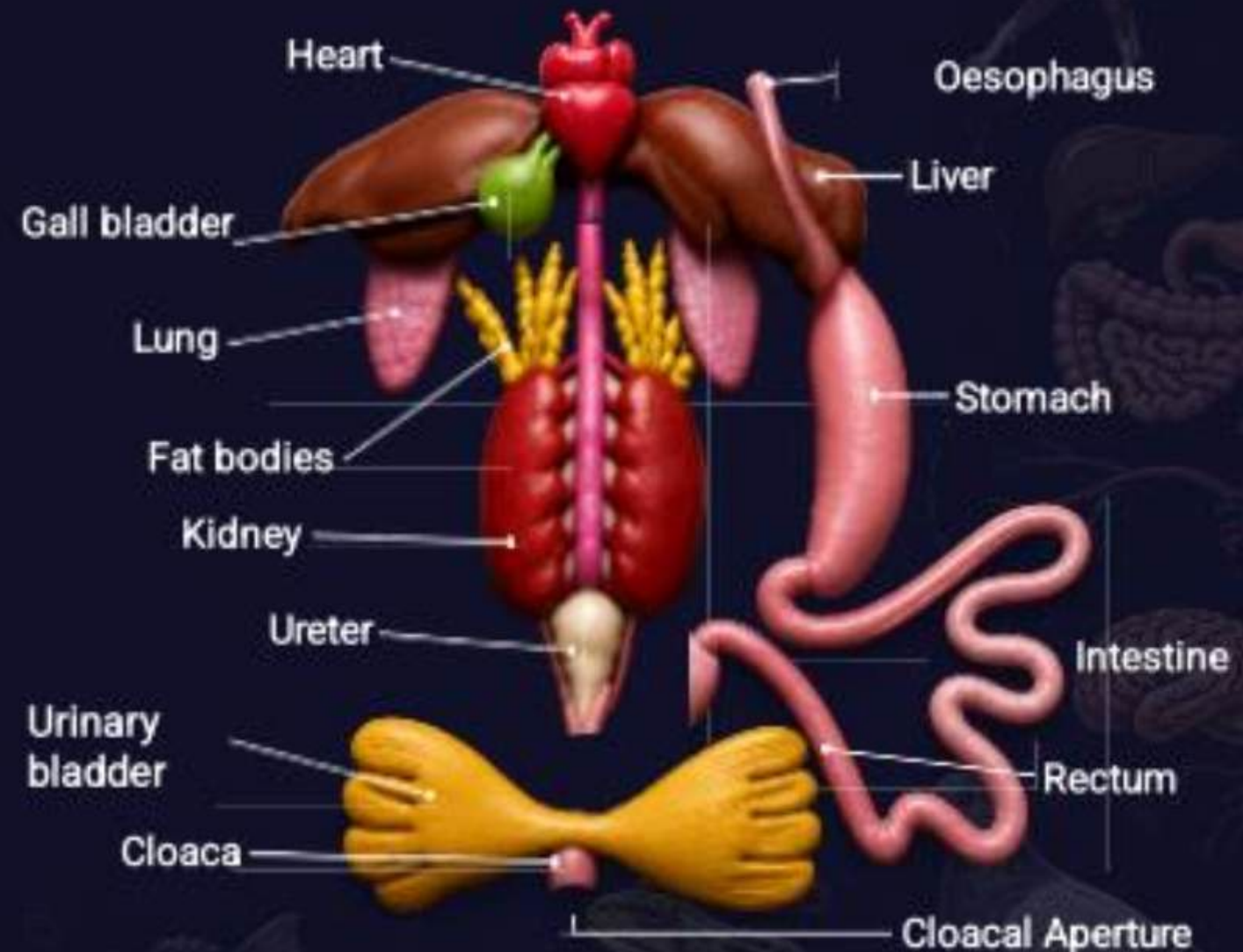
ANATOMY



Different organ systems are accommodated in the frogs' body cavity. Each of these systems have well developed functions and structures.

DIGESTIVE SYSTEM →

It consists of digestive glands and alimentary canal. Frogs, being carnivores, have short alimentary canal. Therefore, their intestinal length is reduced.

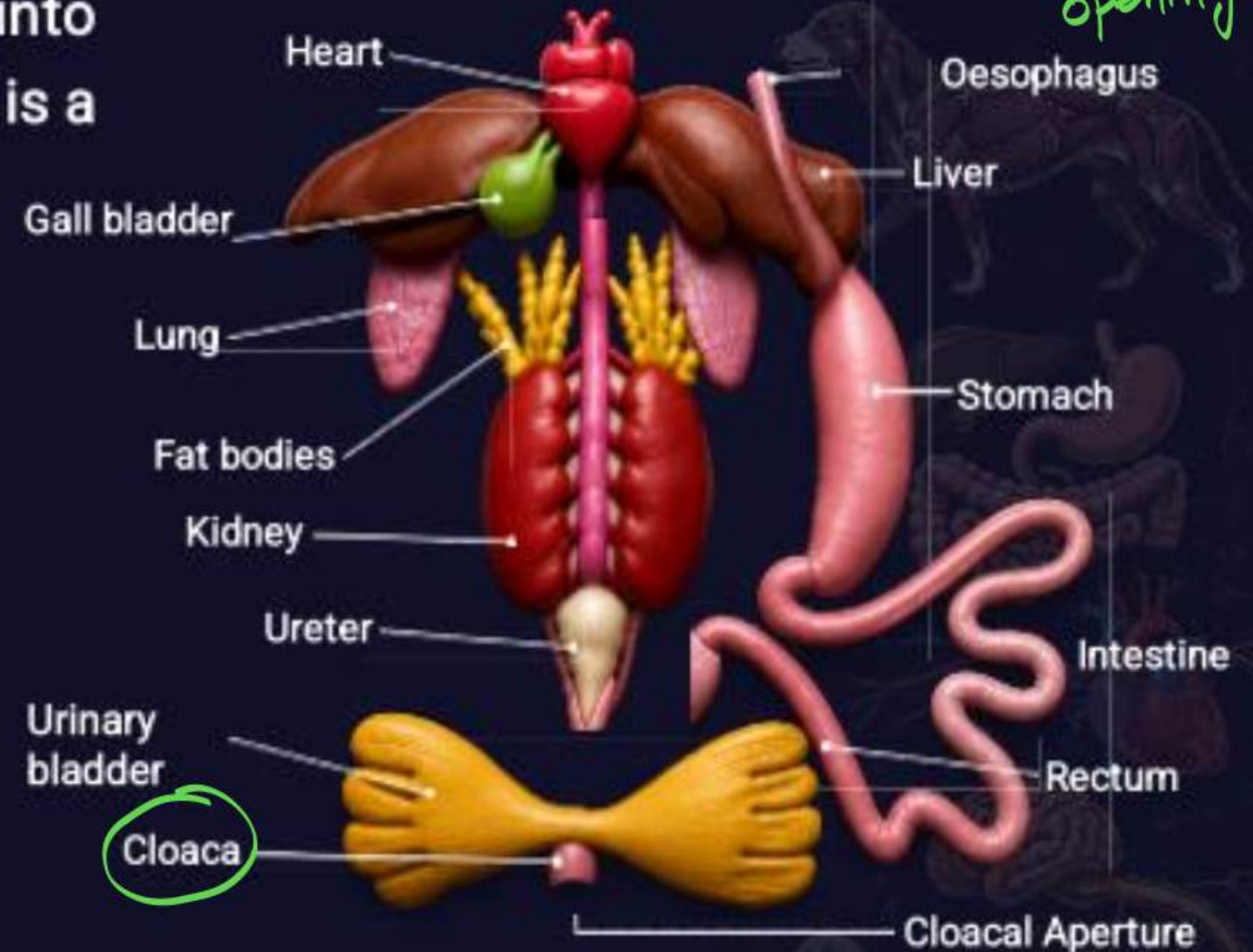


DIGESTIVE SYSTEM

- ✓ Their mouth leads to buccal cavity that opens into pharynx, further leading to oesophagus, which is a short tube.
- Oesophagus continues into the stomach and leads to intestine, then rectum. Cloaca is the final opening to the outside.
- Bile juice is secreted by liver and gall bladder stores it. Besides bile, pancreatic juice is produced by a digestive gland, pancreas. Various digestive enzymes are contained in this pancreatic juice.

Cloaca:- Chamber which Recieves

- ✓ - Alimentary Canal
- ✓ - Urinary Tract
- ✓ - Reproductive Tract

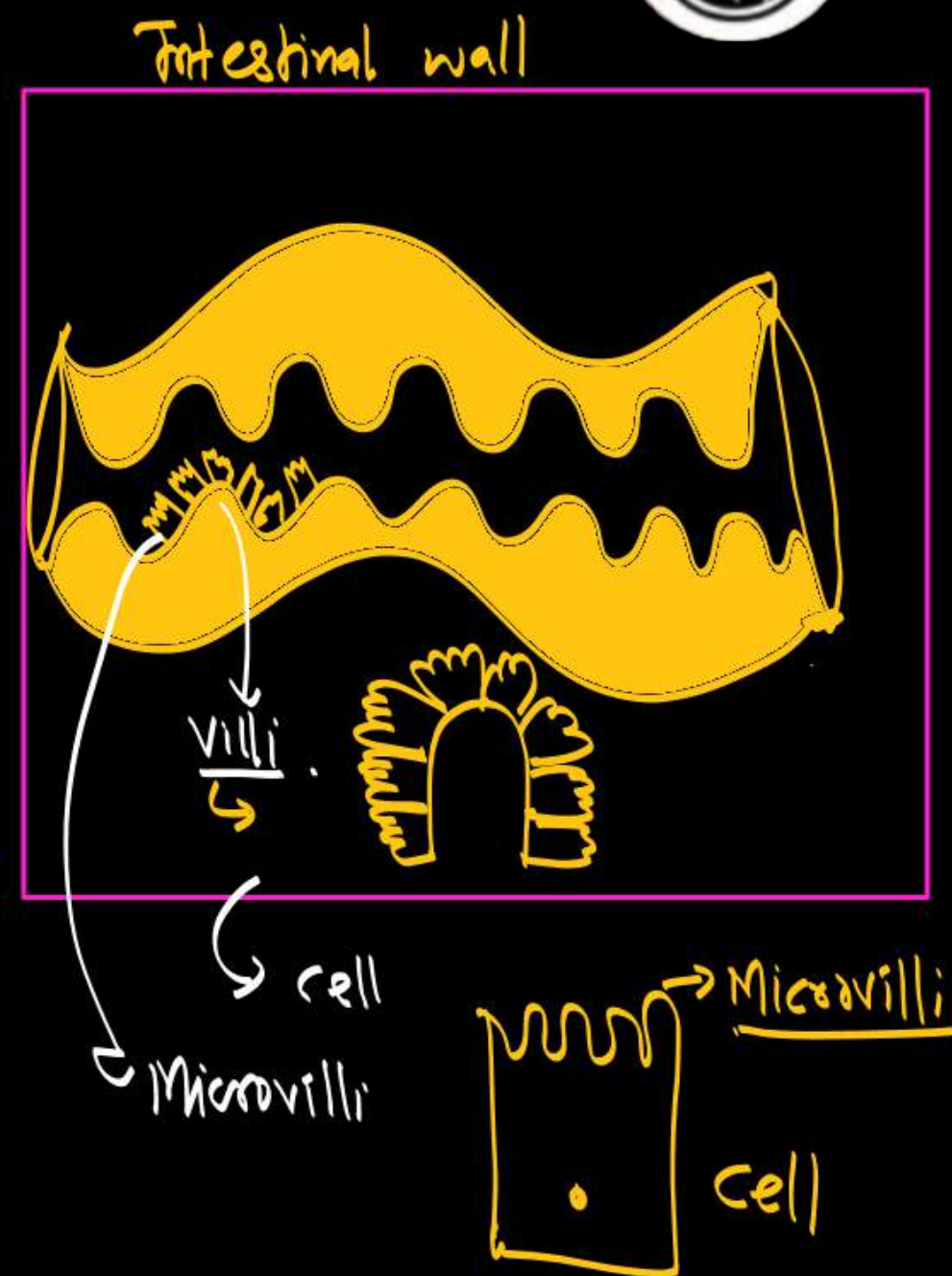
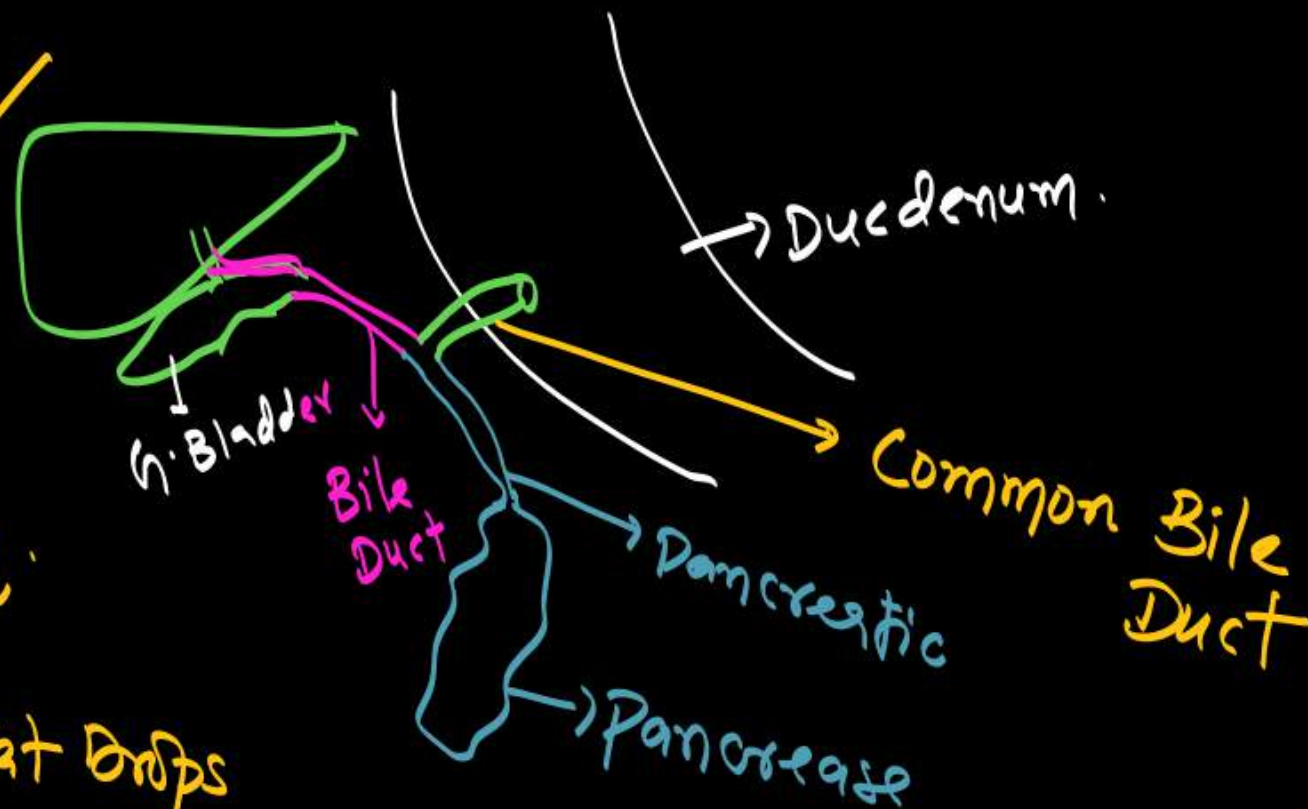


The process of digestion involves the following steps:

1. Bilobed tongue captures the food. (prehensile) → *Used to hold Food/Things*
1. The stomach walls secrete gastric juices and hydrochloric acid whose action cause partial food digestion. This food that is digested partially is called chyme.
1. From stomach, chyme enters duodenum (first part of intestine). Through a common bile duct, pancreatic juice and bile are received here. Fats are emulsified by bile, and proteins and carbohydrates are digested by pancreatic juices. Complete digestion occurs in intestine.
1. Numerous microvilli and villi, that are finger-like folds found in the inner intestinal wall, absorb the digested food.
1. However, the undigested solid waste enters the rectum and through cloaca, the waste is passed out of the body.

Bile

- Emulsification of Fats
- Make Fat Miscible with Watery Content
- By Breaking large Fat Drops in Smaller ones.



RESPIRATORY SYSTEM

Diverse



There are different methods in frogs for respiration. They respire through skin in water and through lungs, skin and buccal cavity on land. The types of respiration include:

Cutaneous respiration or respiration through skin.

Skin functions as respiratory organ in water. Skin also helps in gaseous exchange during hibernation and aestivation on land as well.



RESPIRATORY SYSTEM

Buccal respiration or respiration through **buccal cavity**.

↳ Klas- Buccopharyngeal

Pulmonary respiration or respiration by lungs. Lungs are present in **pair**. They are pink colored, elongated sac-like structures that are found in the upper part of the thorax (trunk region). The entry of air occurs into the buccal cavity through **nostrils**. The air then enters lungs through buccal cavity.

Nostrils
↓
Buccal cavity
↓
Lungs

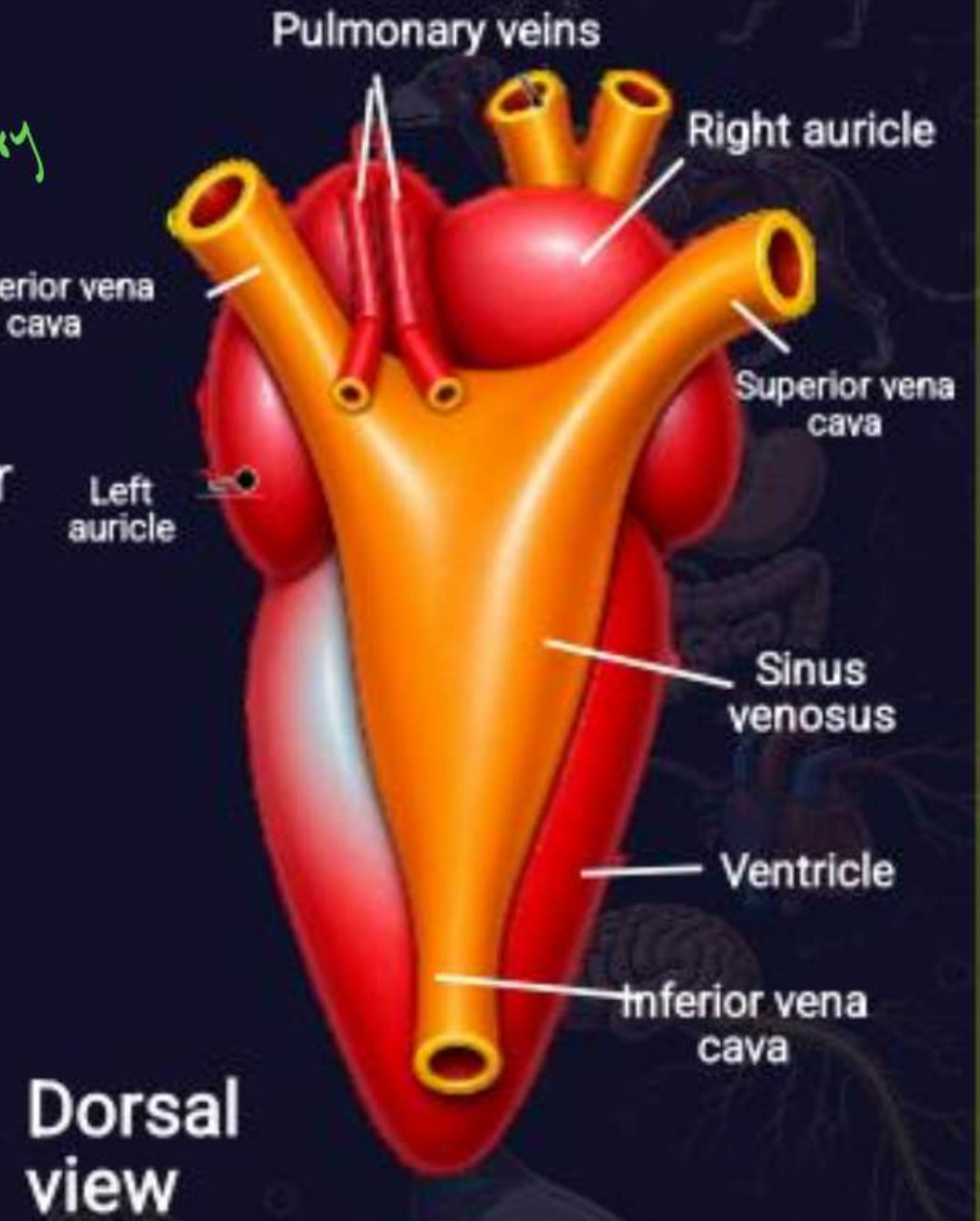
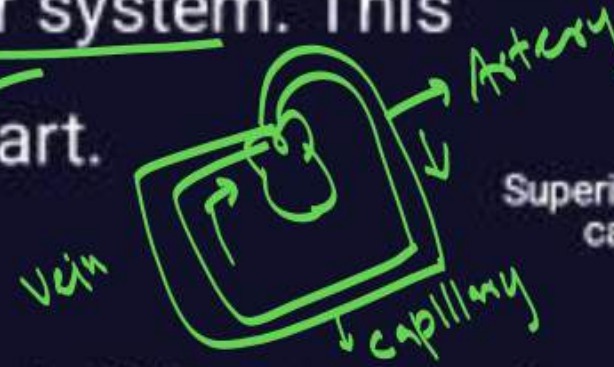


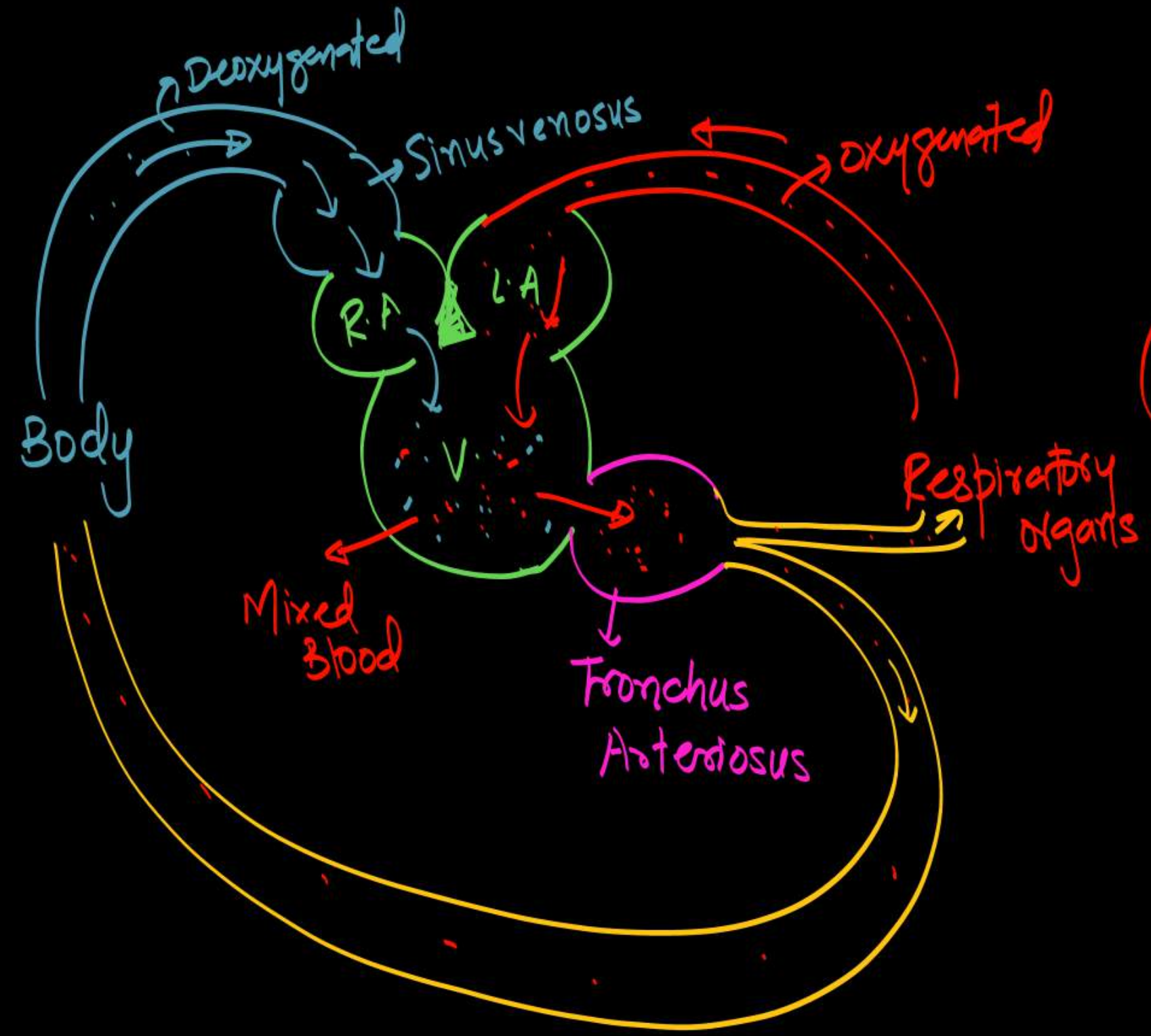
BLOOD VASCULAR SYSTEM



→ Circulatory System

- Frog has closed type well-developed vascular system. This system includes blood, blood vessels and heart.
- Heart is located in the body cavity's upper part. It is a muscular structure.
- Frog's heart is three-chambered having one ventricle and two atria.
- The membrane pericardium covers the heart.



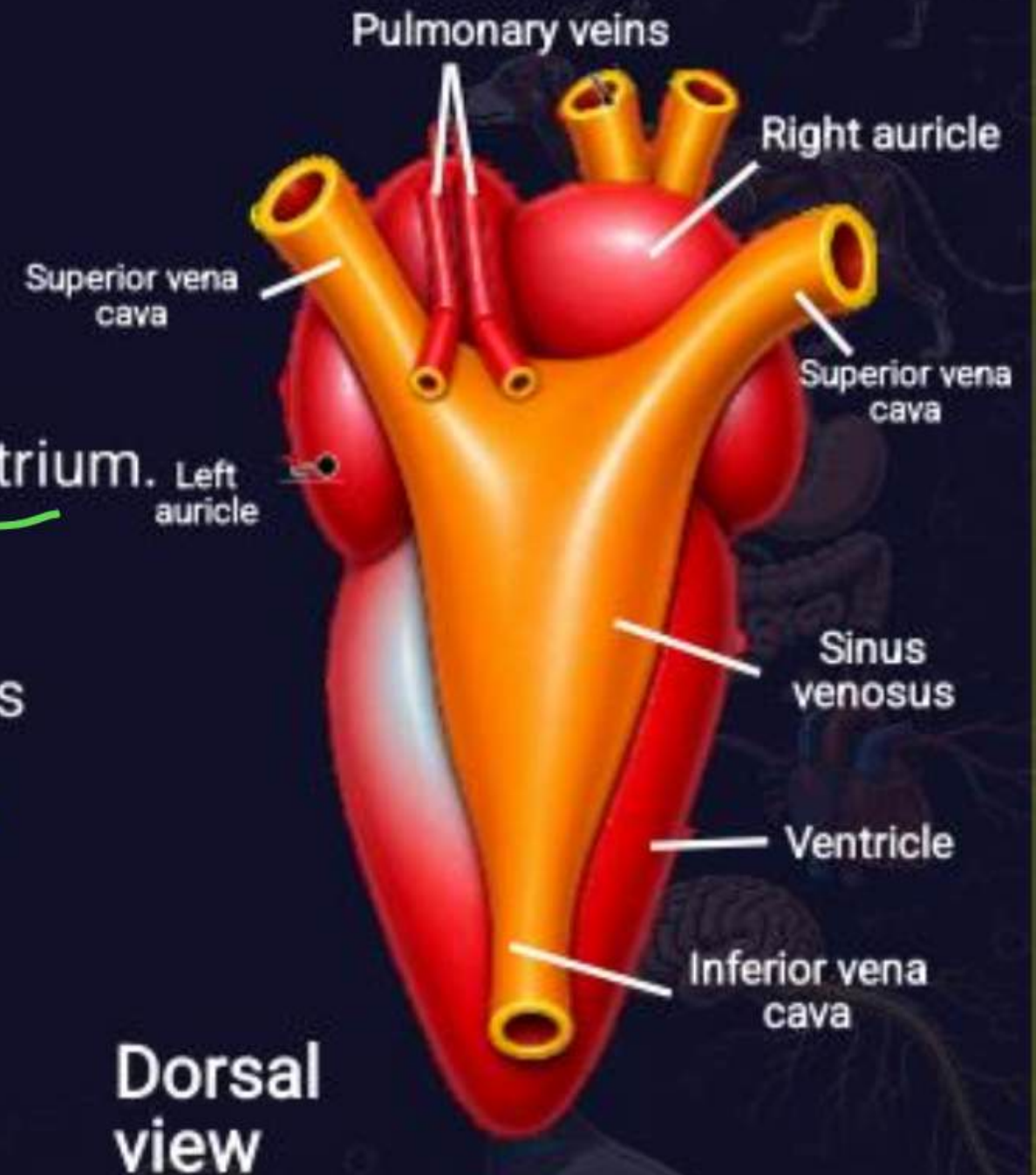


Incomplete Double Circulation

BLOOD VASCULAR SYSTEM



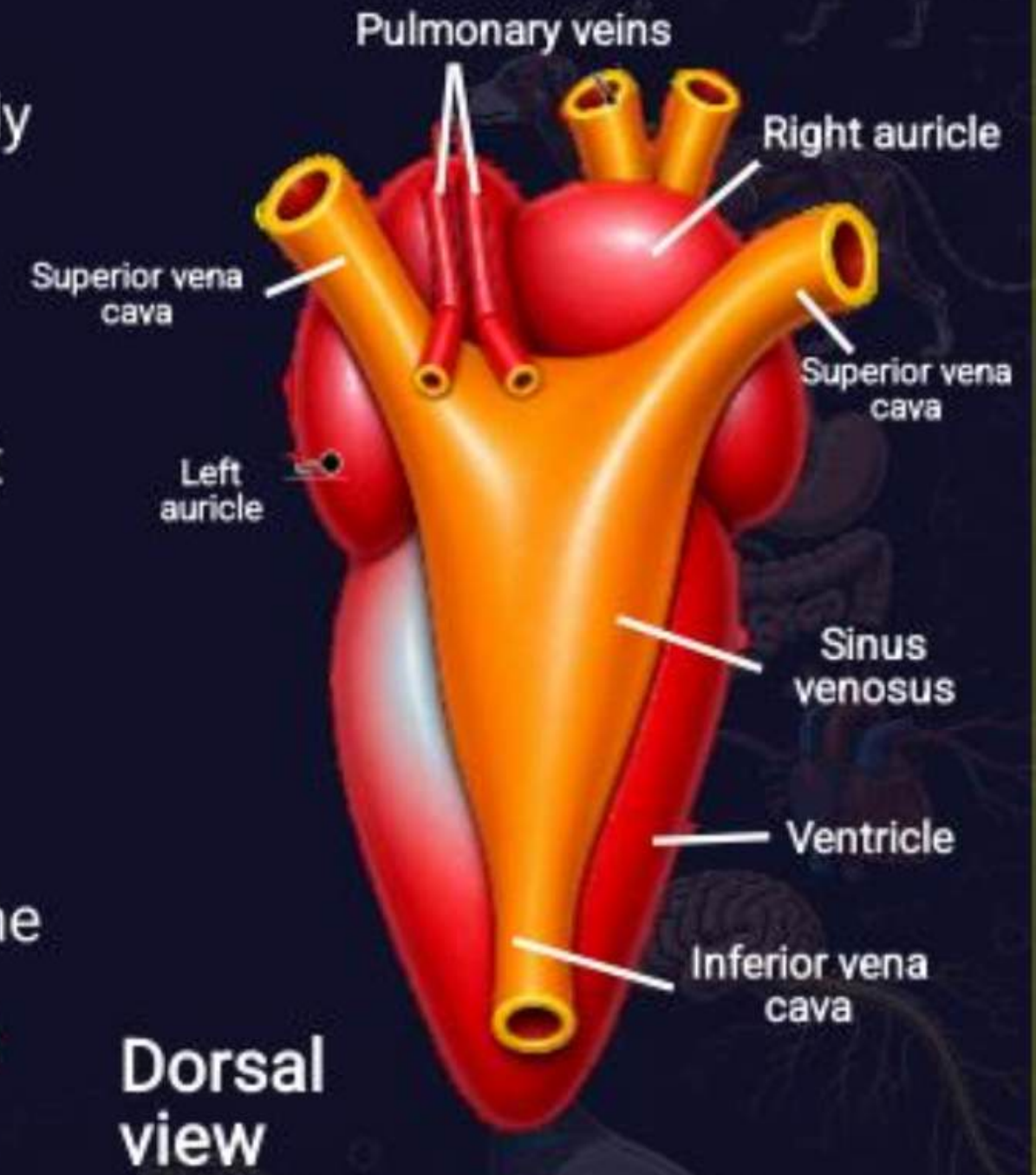
- The right atrium is joined by sinus venosus (a triangular structure). Attached dorsally.
Accessory Chamber
- Vena cava is the major vein that carries blood to the right atrium.
- On the heart's ventral side, the ventricle (mixed blood) opens into conus (truncus) arteriosus, which is saclike. Attached ventrally.



BLOOD VASCULAR SYSTEM



- Arteries form arterial system that carry blood to all the body parts from the heart.
- Veins form venous system that collect blood from different body parts and carry it to the heart.
- In frogs, there is a special venous connection that occurs between the lower body parts and kidney as well as intestine and liver, known as the renal portal system and the hepatic portal system.



Portal System

Body has two Portal System

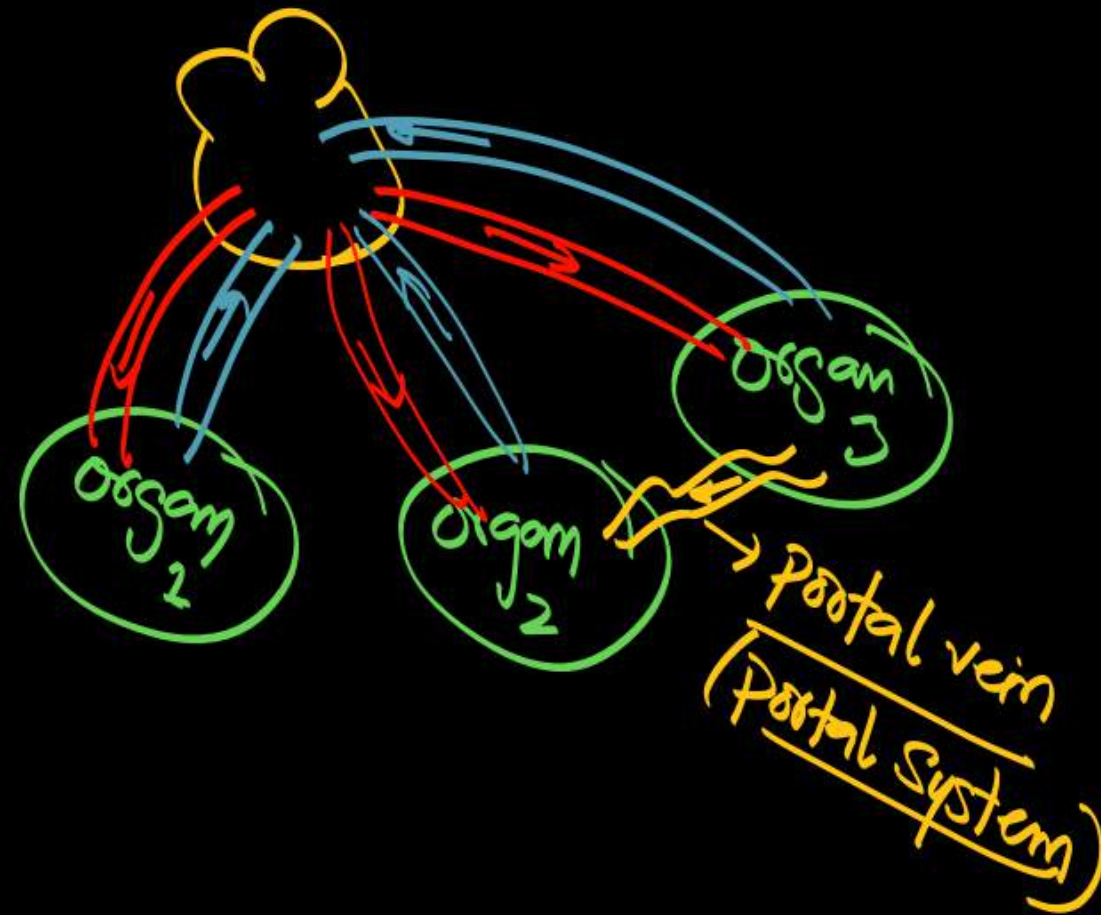
① Renal Portal System

Hind limbs → kidney

② Hepatic Portal System

Intestine → liver

- When body organs are connected by blood vessels and None of them is heart





Q- Frogs have all, except →

(a) → Cloaca

(b) → Trilobed tongue (✓) → Actually Bilobed.

(c) → Renal portal system

(d) → Cutaneous Respiration.

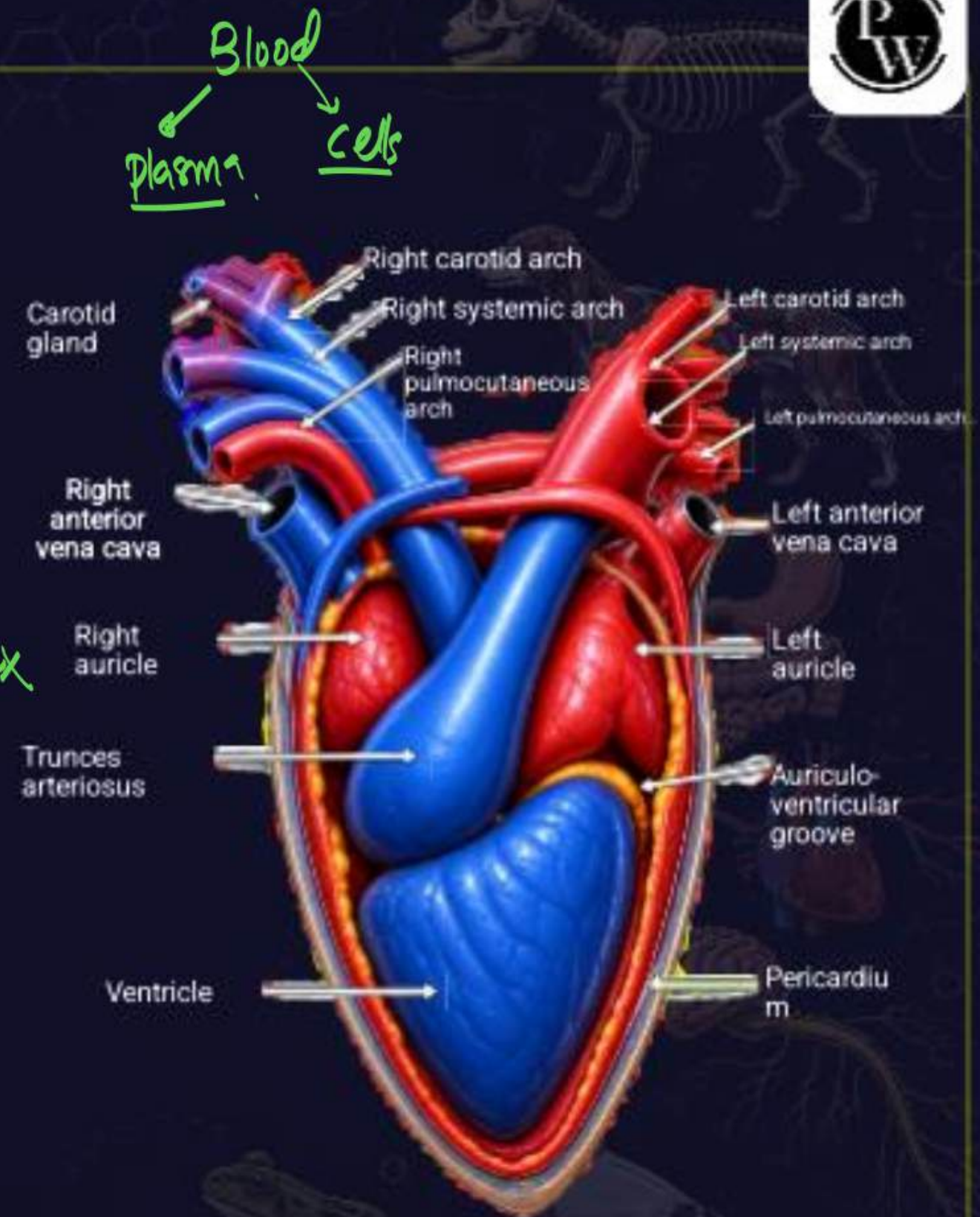
BLOOD VASCULAR SYSTEM



- Blood consists of cells and plasma. Blood cells include platelets, erythrocytes (red blood cells or RBCs), and leucocytes (white blood cells or WBCs). Among these, erythrocytes are nucleated and contain hemoglobin.

Food RBC, = Oval, Nucleated, Biconvex

- ✓ During the circulation, water, gases and nutrients are carried to their respective sites by blood. Heart's pumping action allows the blood circulation.

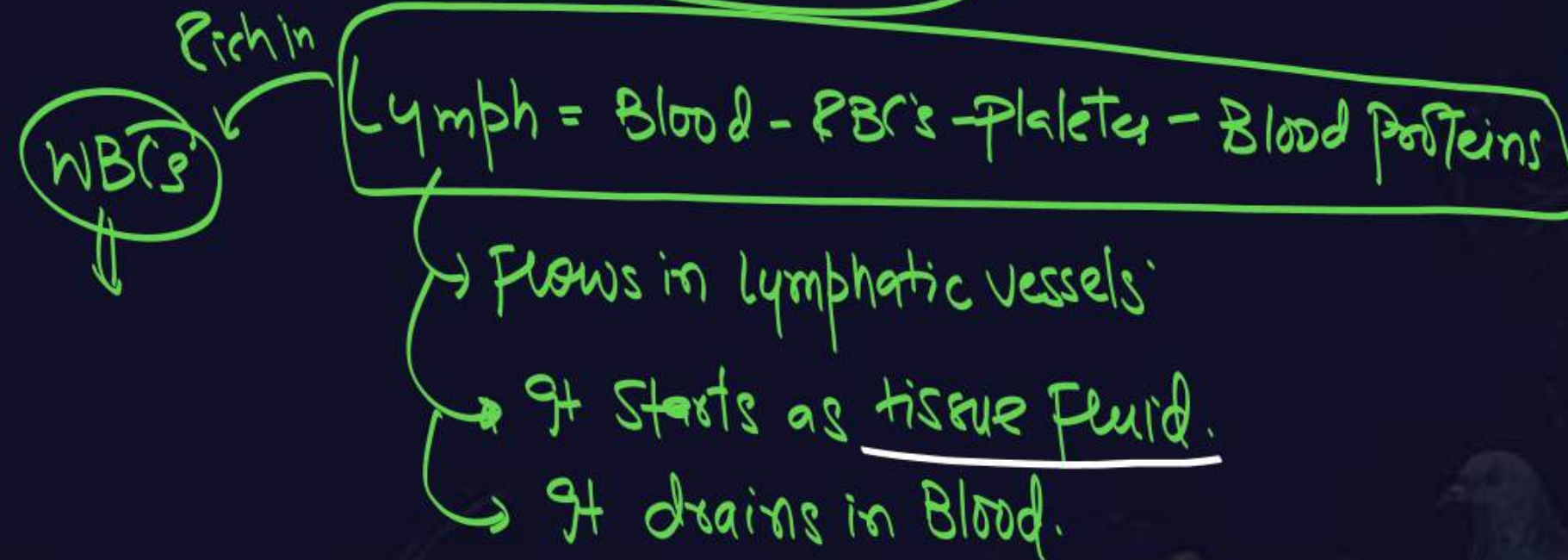


BLOOD VASCULAR SYSTEM



LYMPHATIC SYSTEM

A lymphatic system is also present in frogs. This system includes lymph nodes, lymph channels and lymph. Blood and lymph are different from each other. Lymph do not contain RBCs and a few proteins.



EXCRETORY SYSTEM



Crystal Clear (CC)

- The excretory system in frogs is well developed.

- It helps in nitrogenous wastes (ureotelic) elimination from the body.

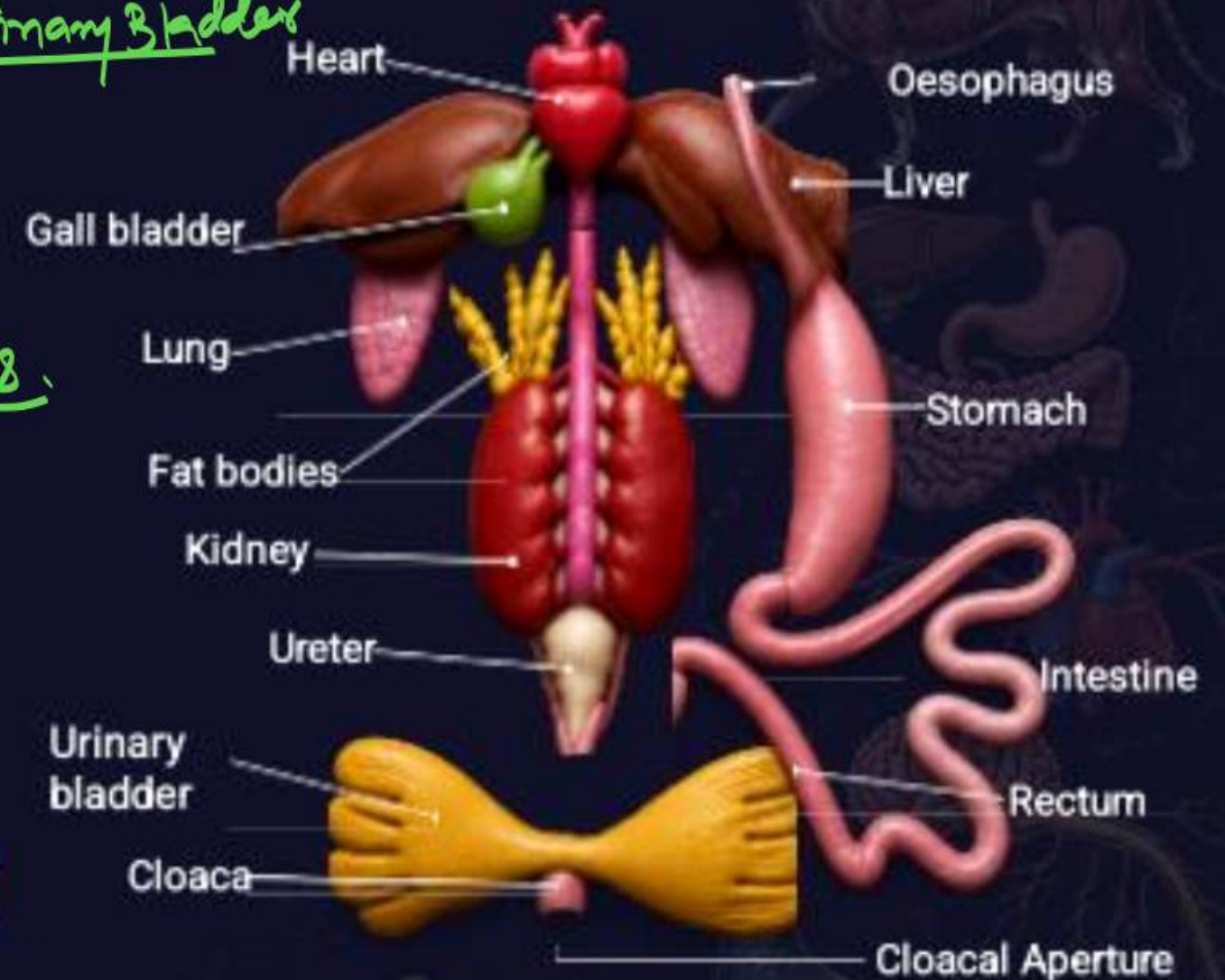
→ frog is Ureotelic like Humans.
Main N₂ Waste = Urea.

- It consists of a kidney pair, urinary bladder, cloaca and ureters.

- ✓ Kidneys are bean like, elongated, dark red, compact structures that are located on both sides of the vertebral column in the body cavity, a little posteriorly.

Lead
Simple

- Kidney
- Ureters
- Cloaca.
- Urinary Bladder



EXCRETORY SYSTEM

- Each kidney consists of several nephrons or uriniferous tubules.

(Class Barchan)

- ✓ Nephrons are the functional and structural units of kidneys.

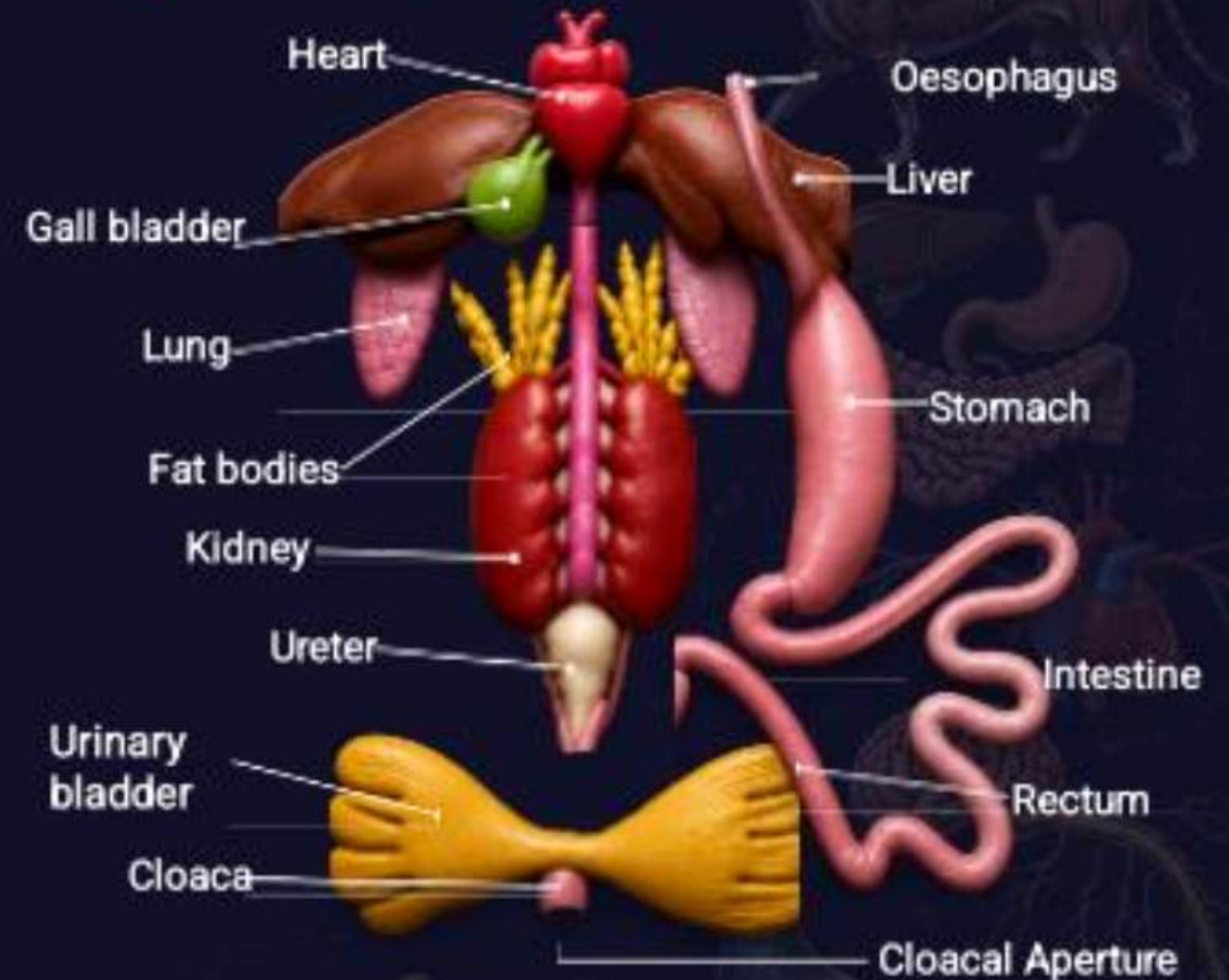
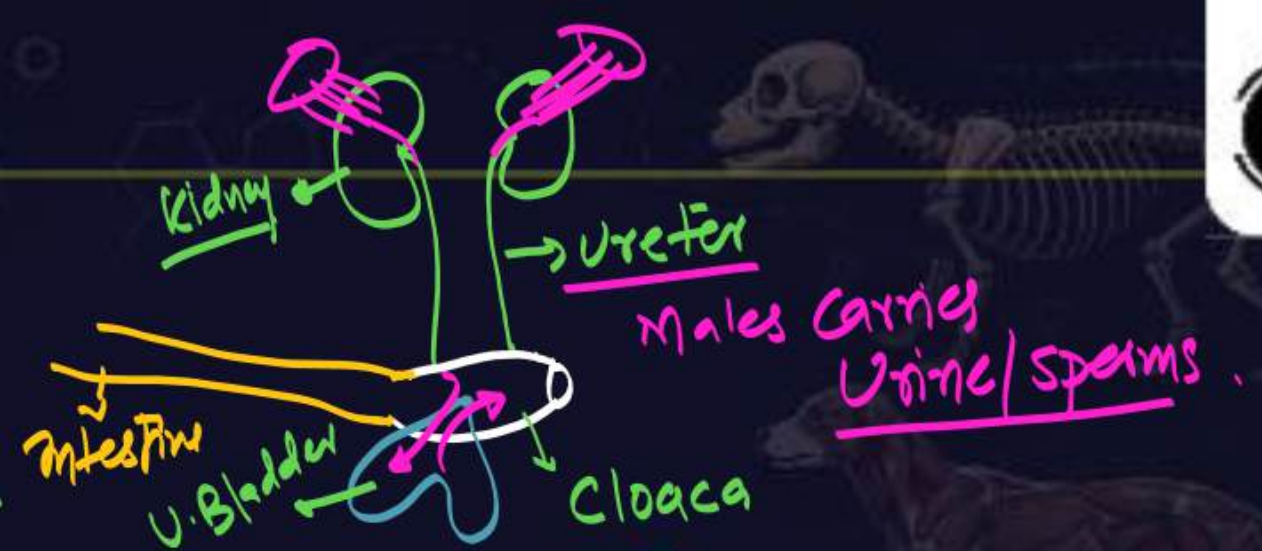
- ✓ In the male frogs, from kidneys emerge two ureters.

- ✓ The ureters opens into the cloaca by functioning as

urinogenital duct.

Urine

Semen (Sperms)

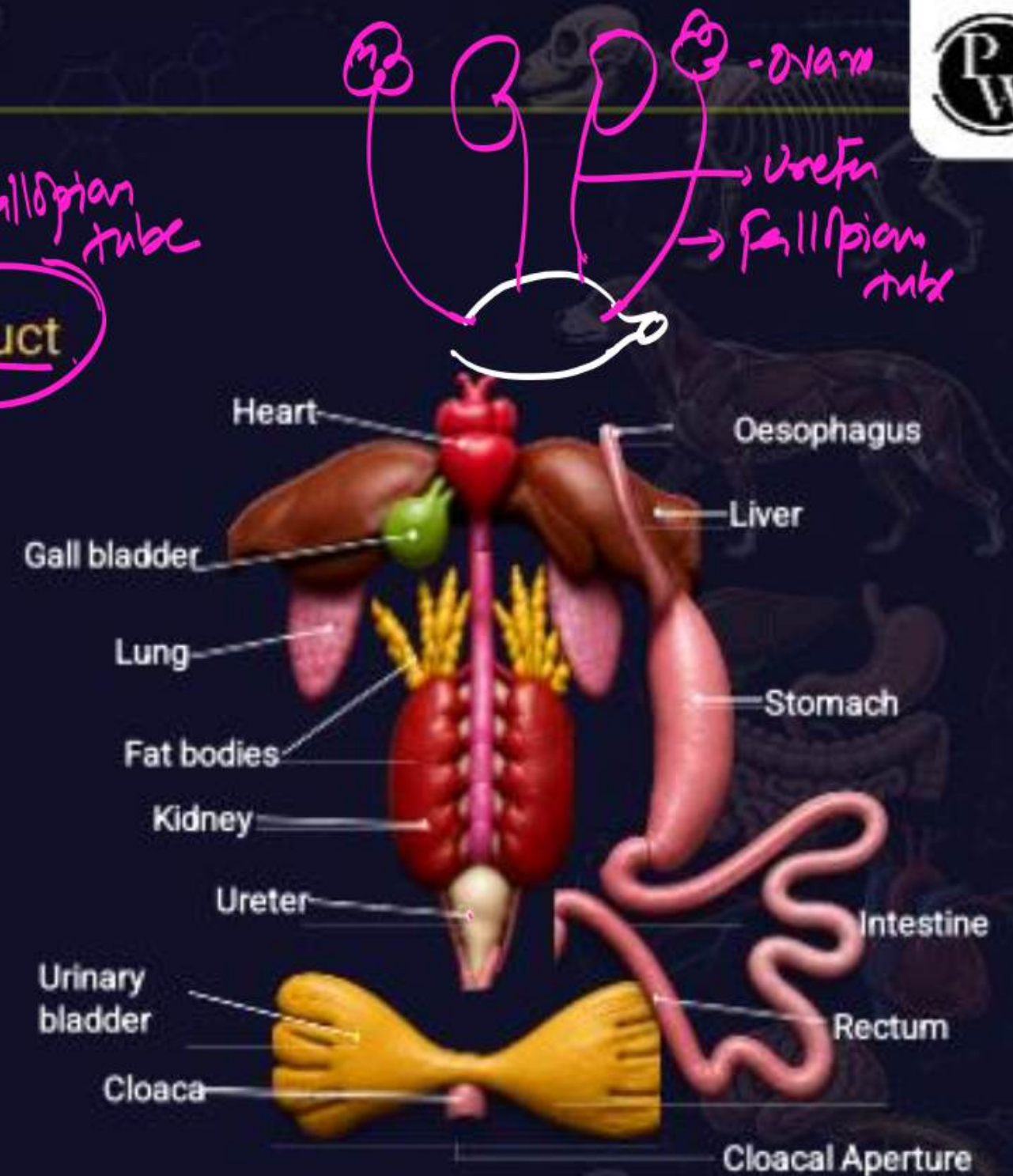


EXCRETORY SYSTEM

- However, in females, there is separate opening of the oviduct and ureters in the cloaca.

✓ The urinary bladder is thin-walled and is found to the rectum's ventral side. It continues into the cloaca.

- Frogs are ureotelic animals as urea is excreted by them. Blood carries excretory wastes into the kidney.
- The separation and excretion of these wastes occur in the kidney.

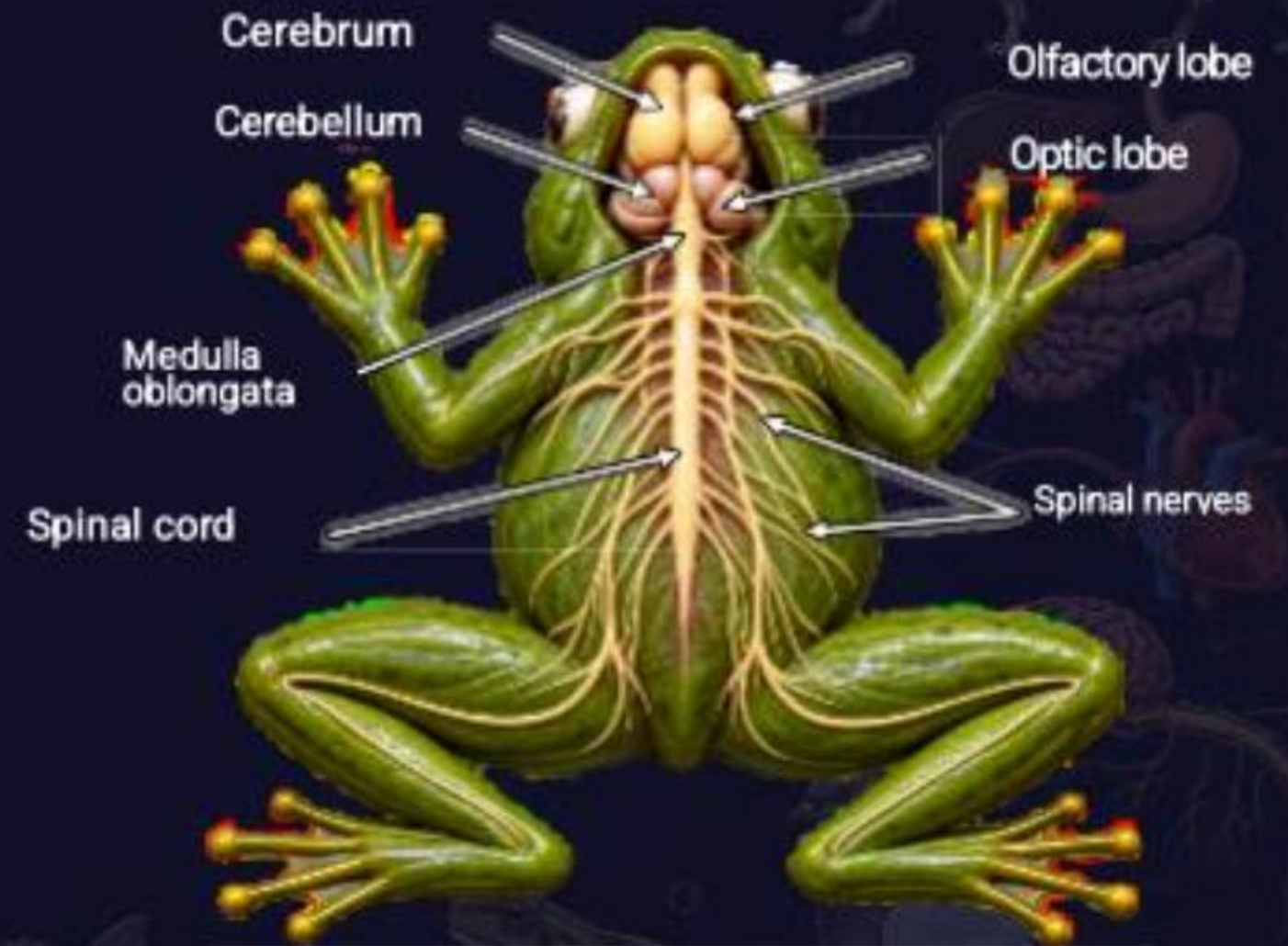


Nervous System



The system for control and coordination is highly evolved in the frog. It includes both **neural system** and **endocrine glands** (*Harmonal system*)

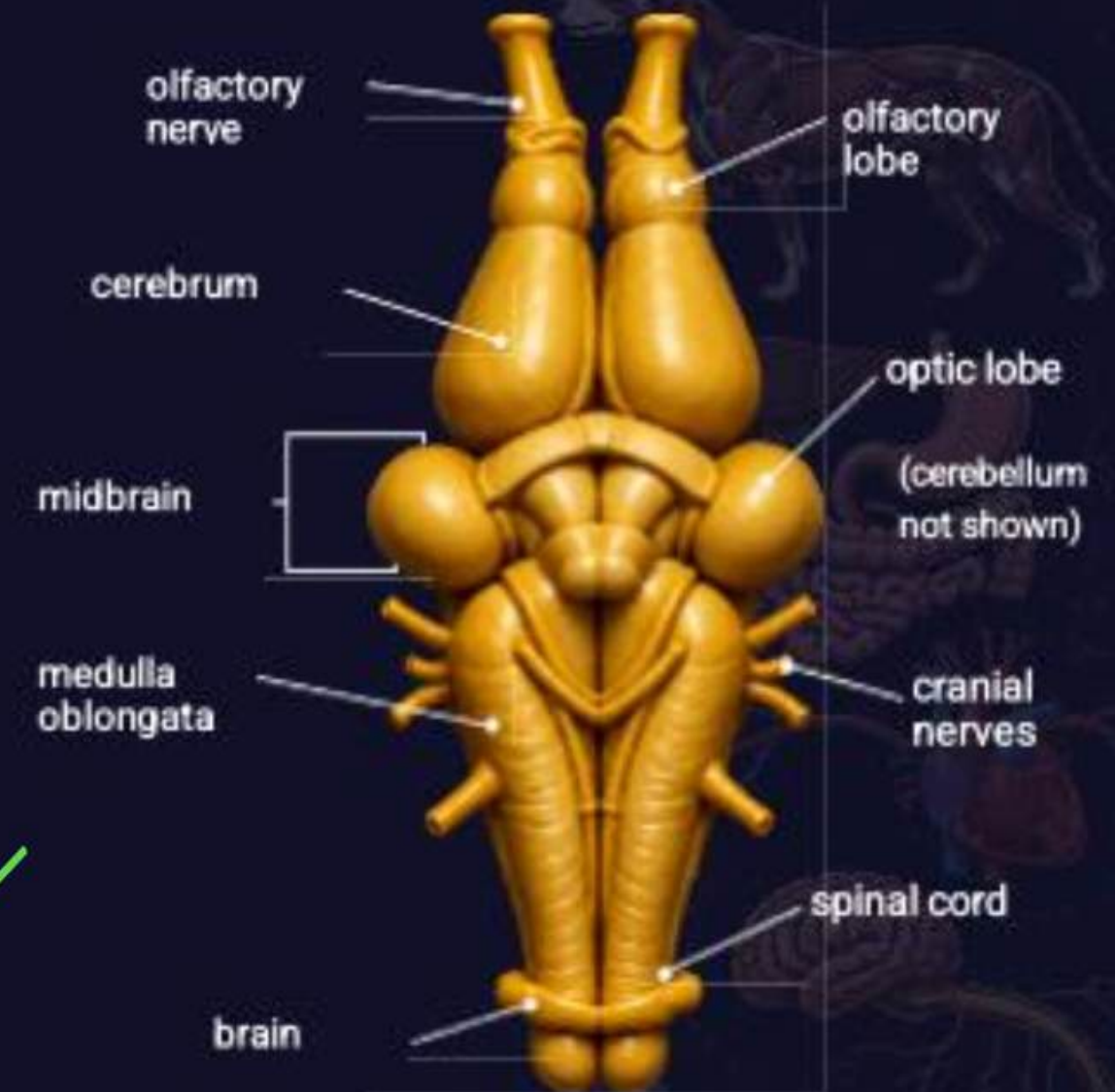
Endocrine System – The endocrine glands secrete various **hormones** that help in **coordination of various body organs** chemically. In frogs, the endocrine glands that are prominent include **thyroid**, **thymus**, **pancreatic islets**, **gonads**, **pituitary**, **parathyroid**, **pineal body**, and **adrenals**

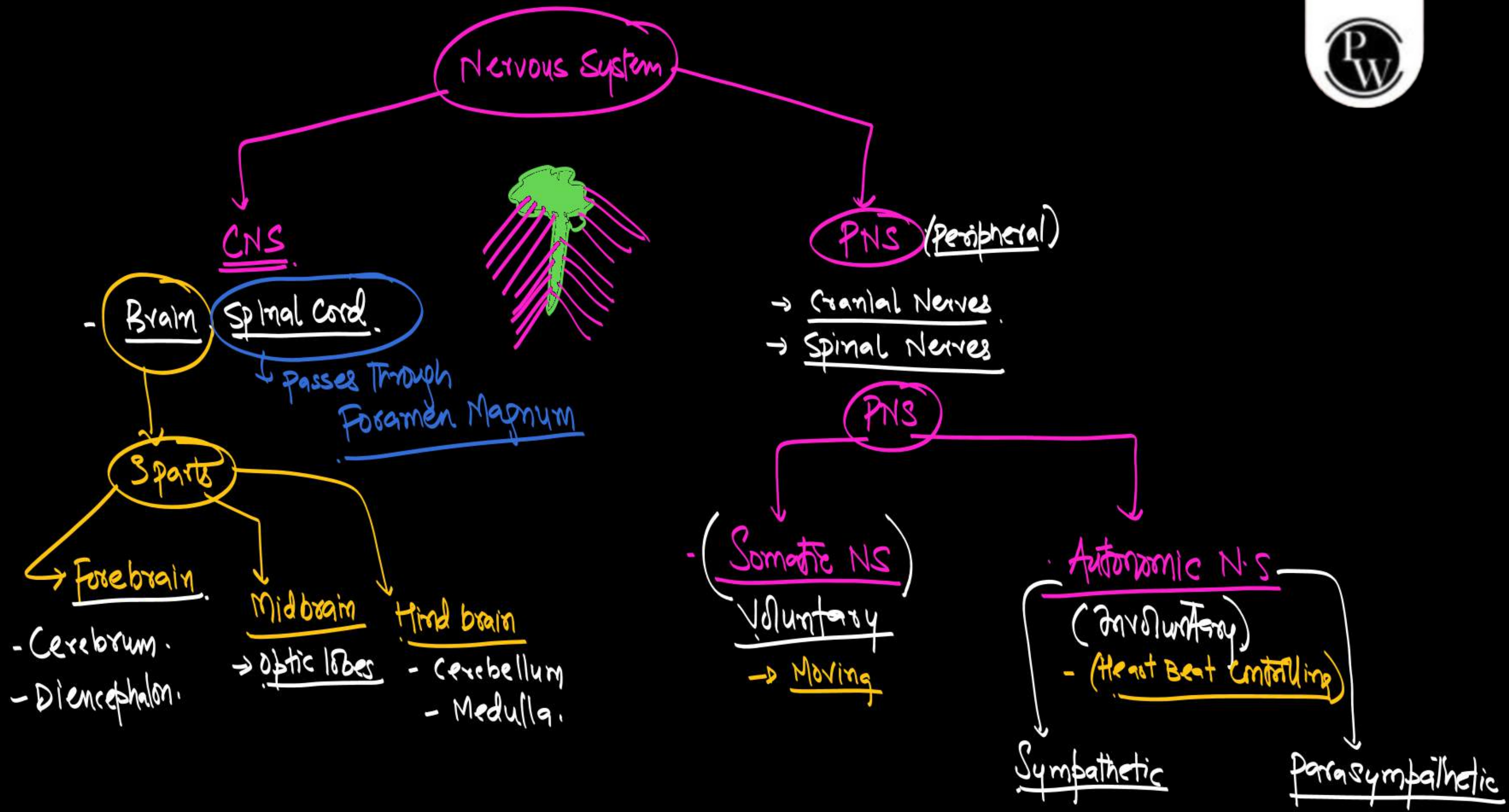


Nervous System

Human - 12 pairs

Nervous System – This system is organized into a central nervous system (brain and spinal cord), a peripheral nervous system (cranial and spinal nerves) and an autonomic nervous system (sympathetic and parasympathetic). There are ten pairs of cranial nerves arising from the brain. Brain is enclosed in a bony structure called **brain box (cranium)**. There are three parts of the brain – **forebrain, midbrain and hindbrain**. ✓





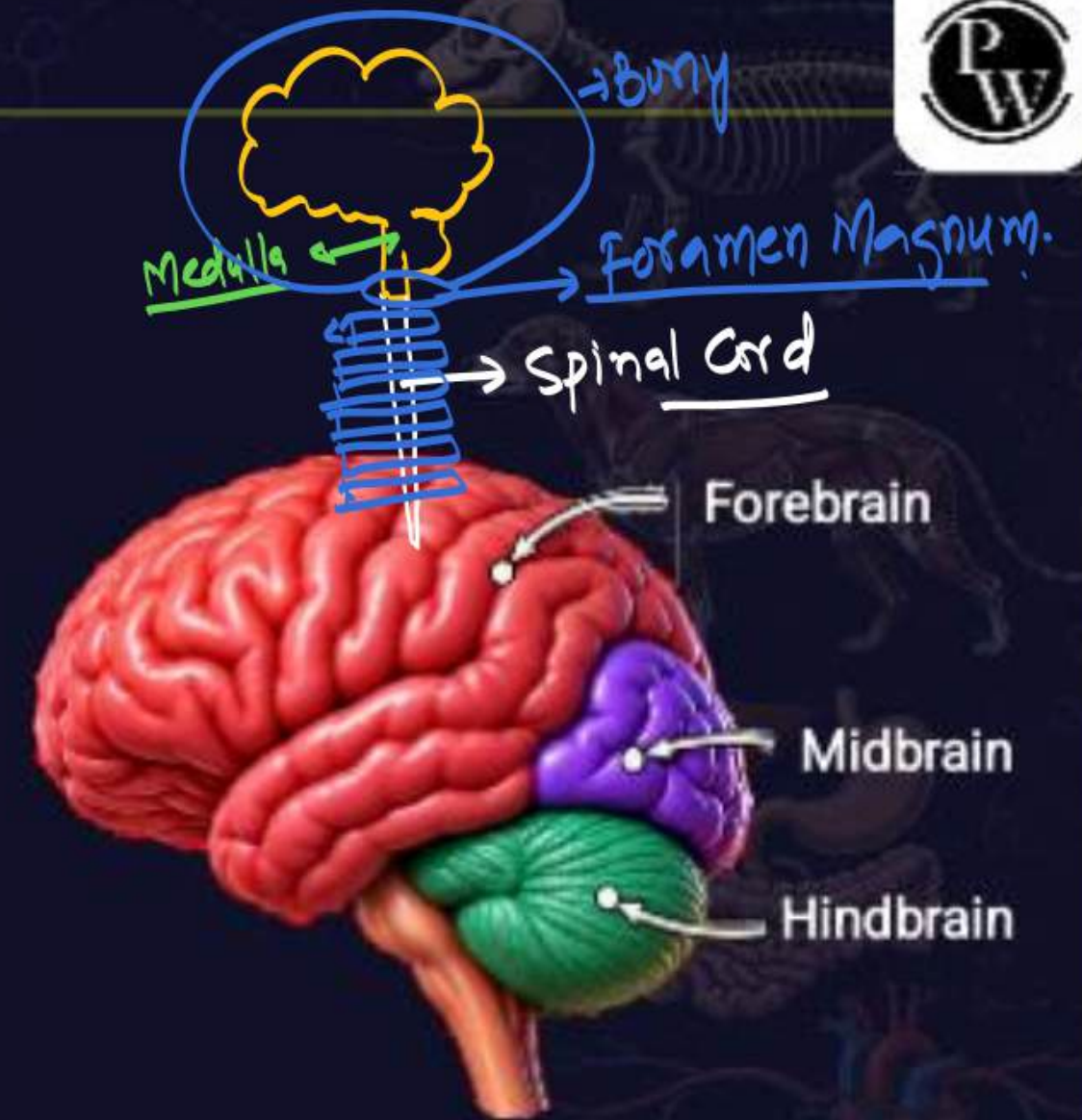
Nervous System



Forebrain – includes olfactory lobes, paired cerebral hemispheres and unpaired diencephalon

Midbrain is characterized by a pair of optic lobes.

- ✓ **Hindbrain** consists of cerebellum and medulla oblongata.
- ✓ The medulla oblongata passes out through the foramen magnum and continues into spinal cord, which is enclosed in the vertebral column.



Nervous System



Frog has different types of sense organs ✓ (5)

- organs of touch (sensory papillae) ✓
- taste (taste buds) ✓
- smell (nasal epithelium) ✓
- vision (eyes) ✓
- hearing and balancing / equilibrium (tympanum with internal ears).

Ear - organ of Hearing, Balancing



Nervous System



Eyes and internal ears are well-organized structures and the rest are cellular aggregations around nerve endings.

Their eyes are present in skull's orbit in pair and are spherical structures. They possess single unit and hence are simple eyes.

The prominent endocrine glands found in frog are pituitary, thyroid, parathyroid, thymus, pineal body, pancreatic islets, adrenals and gonads.



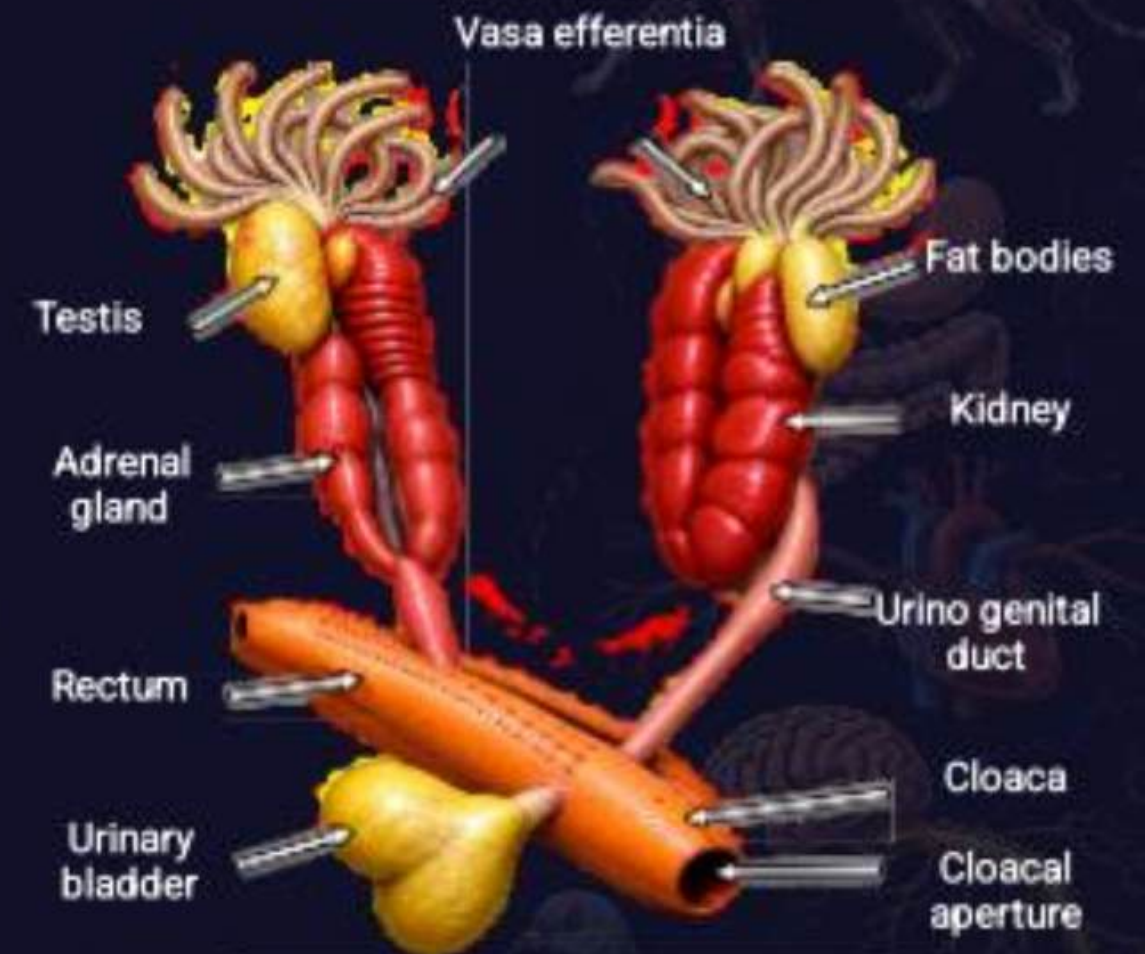
REPRODUCTIVE SYSTEM



In frogs, female and male reproductive systems are well organized

MALE REPRODUCTIVE SYSTEM

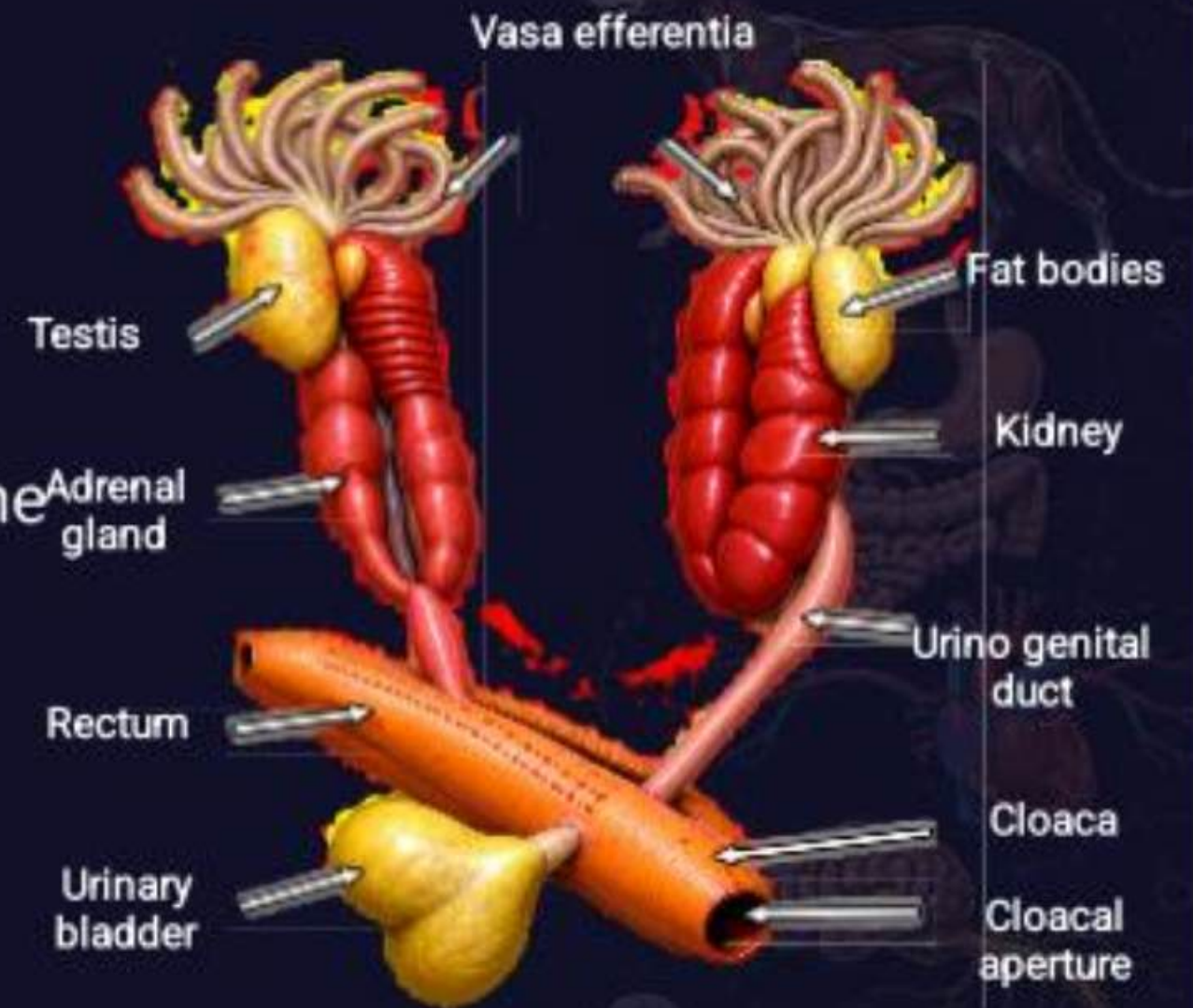
- It consists of a pair of testes that are ovoid, yellowish and are adhered by mesorchium (a double fold of peritoneum) to the kidneys' upper part.
- From testes, arise vasa efferentia that are 10 – 12 in number.
- They open into Bidder's canal and enter on the side of the kidneys.



REPRODUCTIVE SYSTEM

- This is followed by its communication with the **urinogenital duct** that leaves the kidneys with its further opening into the **cloaca**.

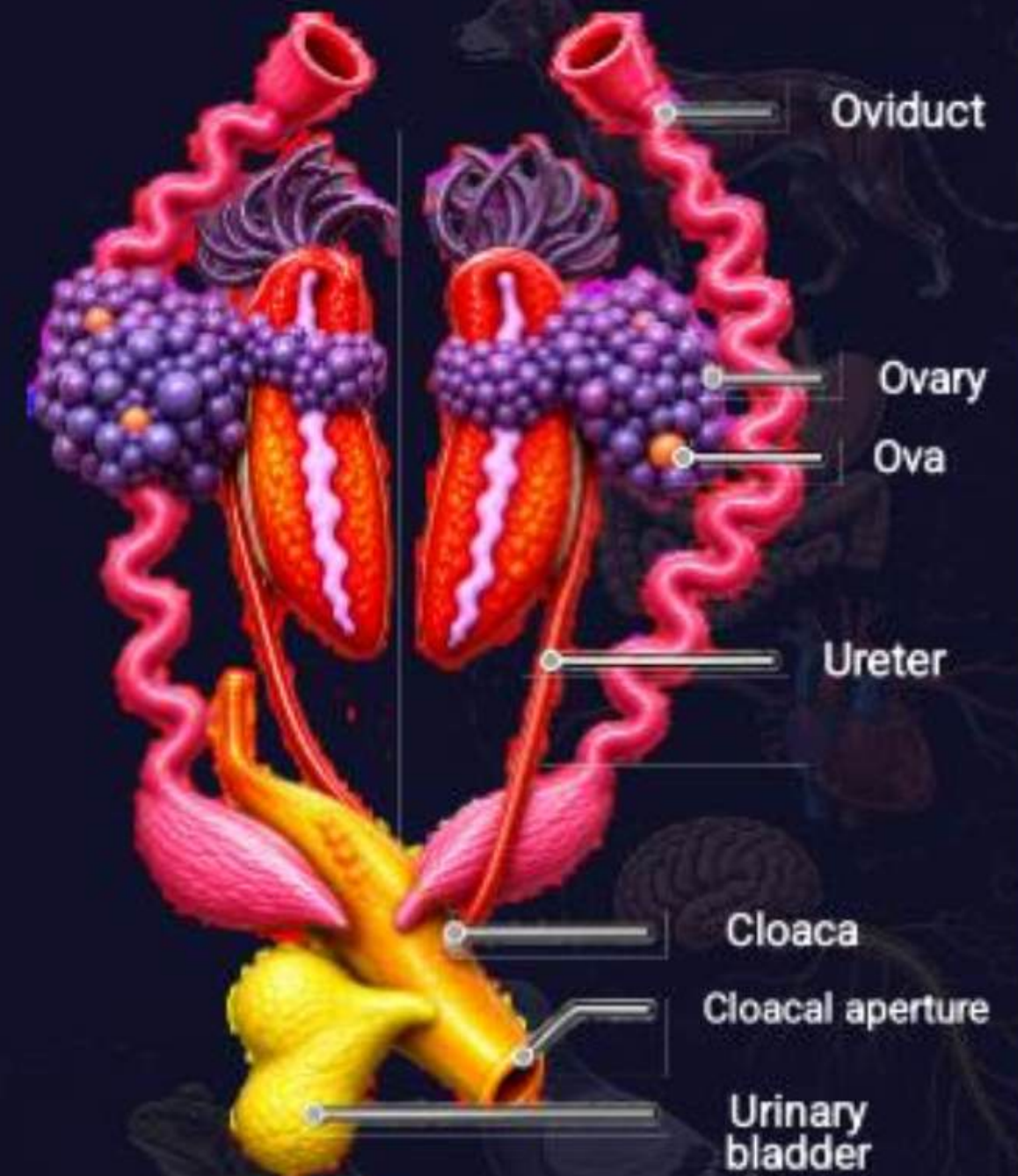
- The cloaca is a median, small chamber that allows the passage of **sperms, urine and faecal matter** to the outside.



REPRODUCTIVE SYSTEM

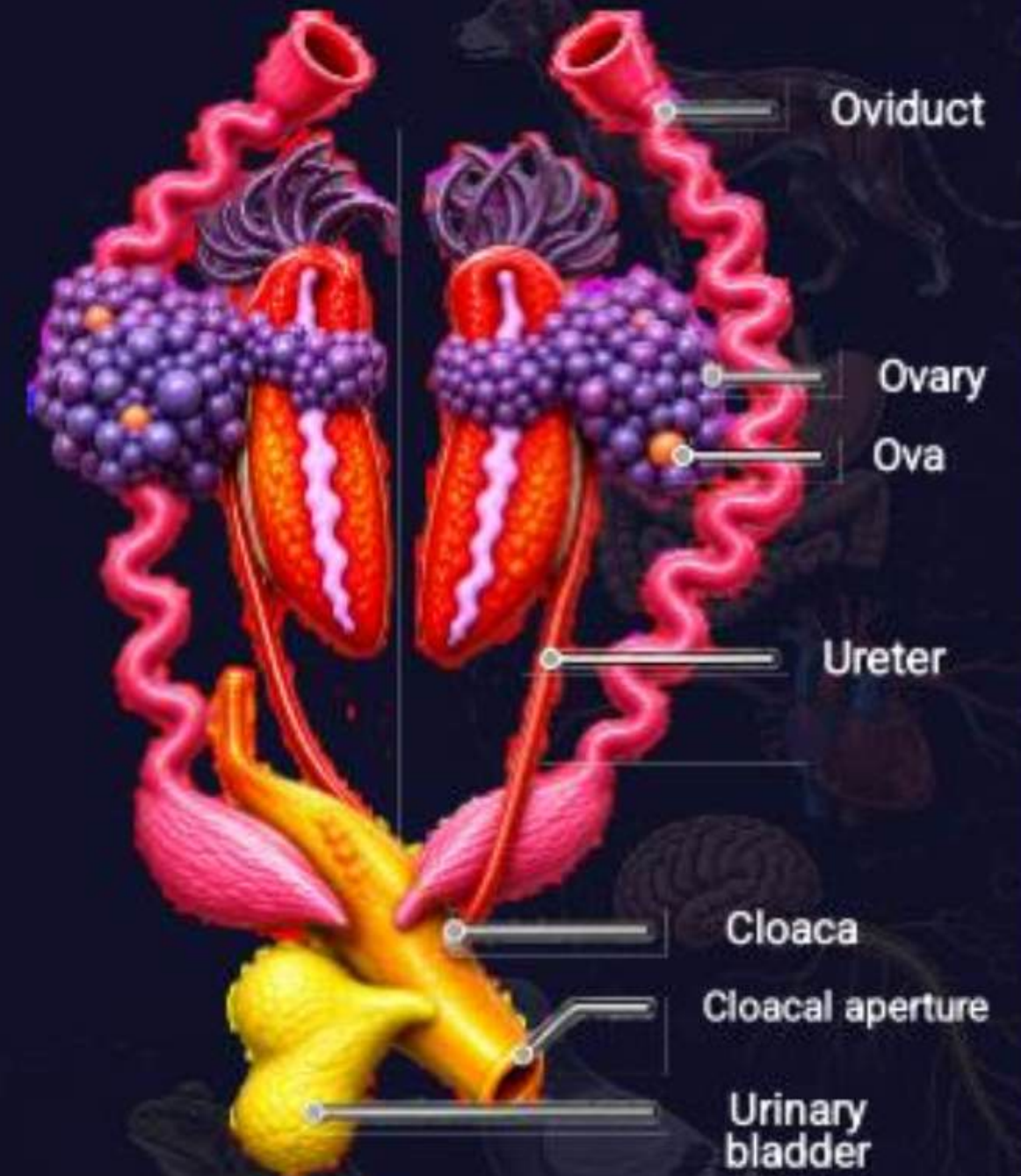
FEMALE REPRODUCTIVE SYSTEM

- It consists of a pair of ovaries that are found near kidneys, but are not functionally connected with them.
- From the ovaries arise a pair of oviduct that opens separately into the cloaca.
- At a time, 2500 – 3000 ova can be laid by a mature female.



REPRODUCTIVE SYSTEM

- Frogs show **external** fertilization that occurs in water. This is followed by the development into tadpole, which is the larval stage.
- Finally, an adult is formed by the **metamorphosis** of the tadpole.



ECOLOGICAL IMPORTANCE OF FROG

- Frogs are quite beneficial as they provide protection to the crop and eat insects
- In the ecosystem, they provide an important link between the food chain and food web, thereby maintaining the ecological balance.
- Frogs muscular legs are eaten by man in some countries.



Thank You